

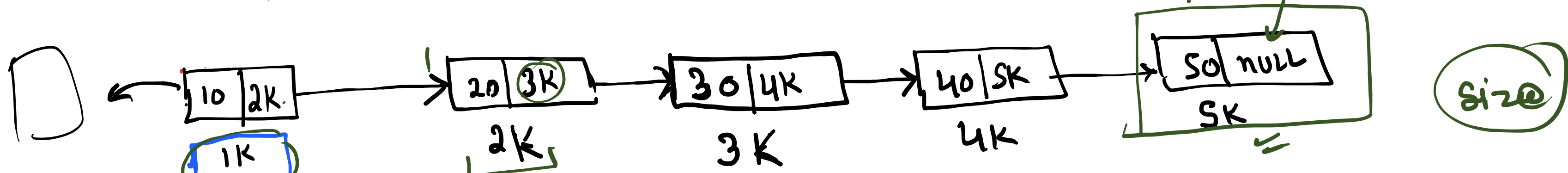
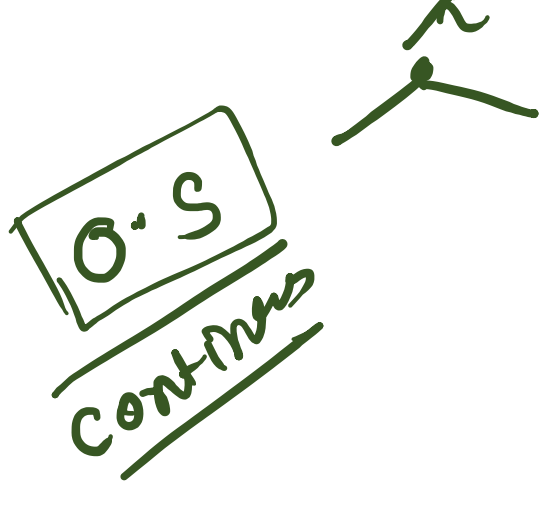
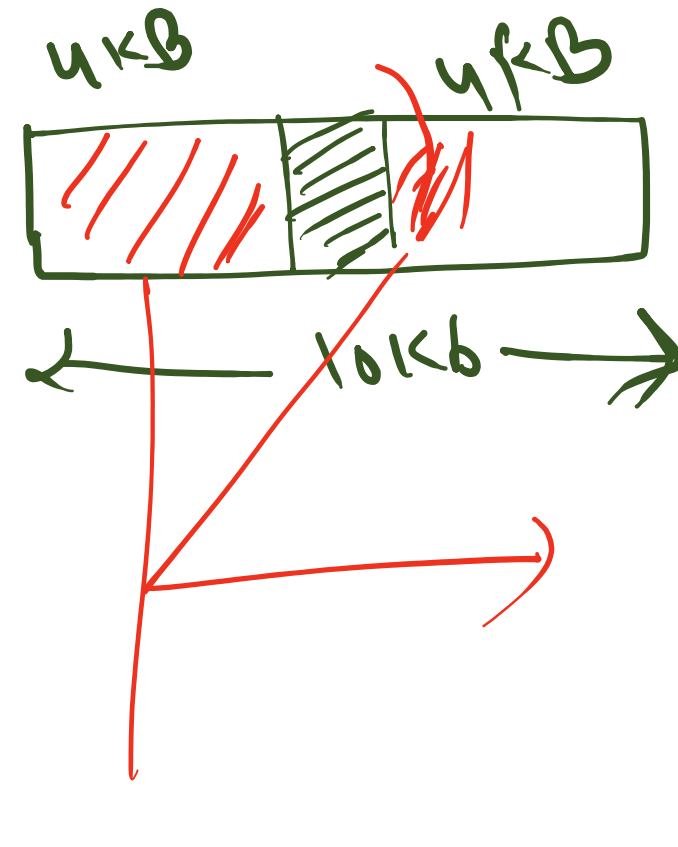
LinkedList

non-continuous
Lx

Array

5
20 byte

P1 = 6K



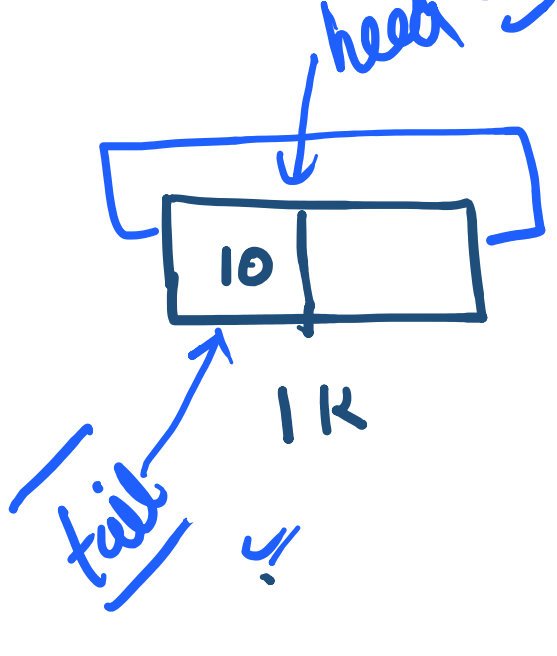
```
class Node {
    int val;
    Node next;
}
```

```
class Student {
    String name;
    int age;
}
```

Student s = new Student();

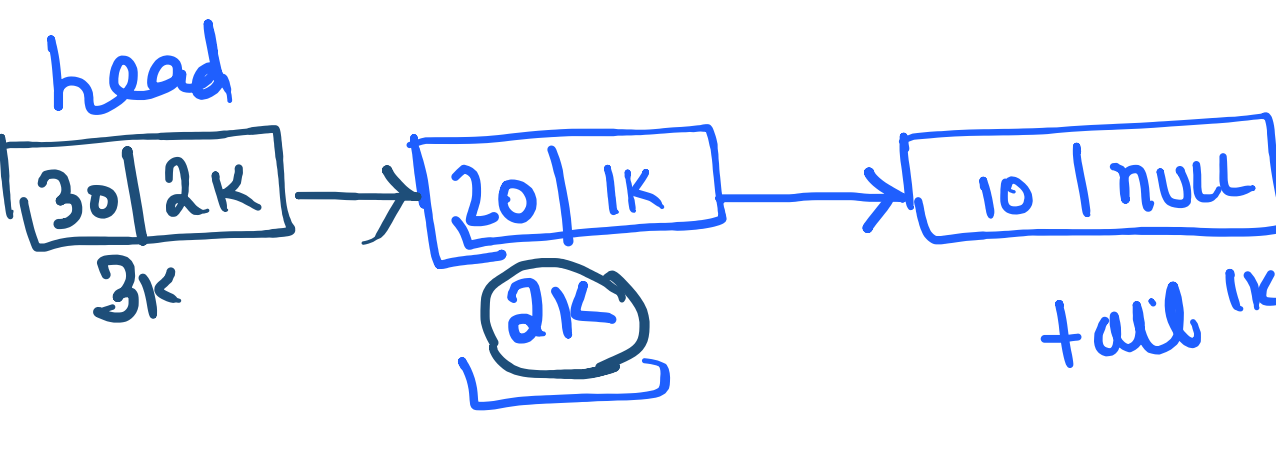
name: age: date: Address:

addfirst
addlast
add at index
remove first
remove last
remove at index
get first
set last
set at index
display

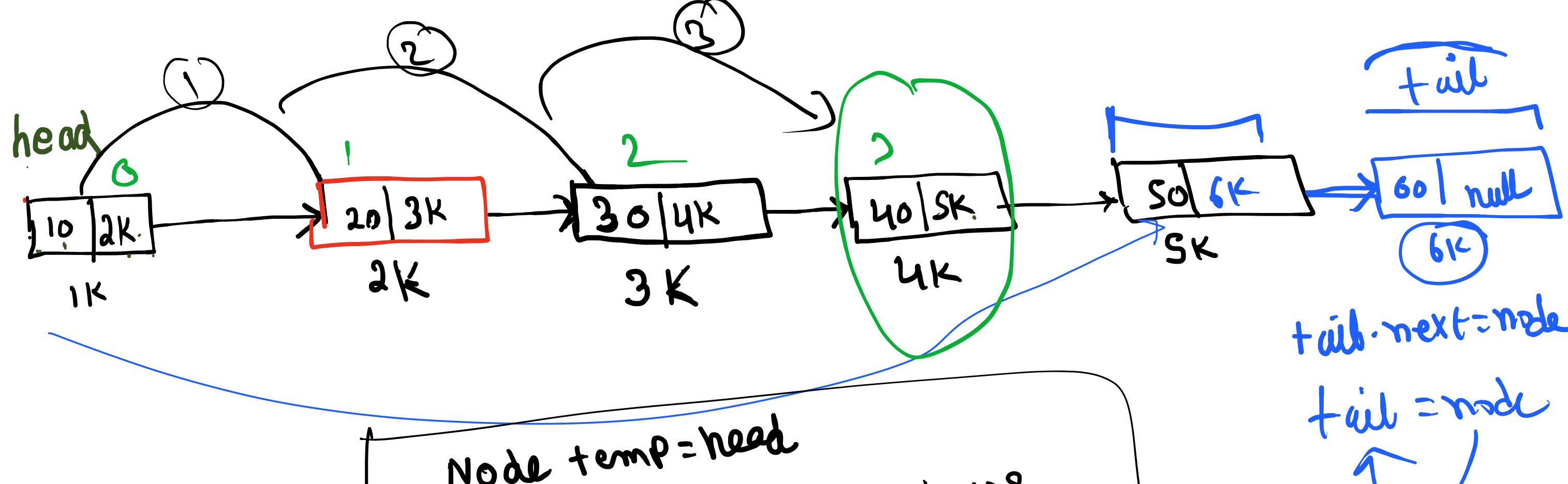


addfcioj
Node -> 1K

```
public void addfirst(int item) {
    Node node = new Node(item);
    if (size == 0) {
        head = node;
        tail = node;
        size++;
    } else {
        node.next = head;
        head = node;
        size++;
    }
}
```



addfcioj
addfcioj
addfcioj



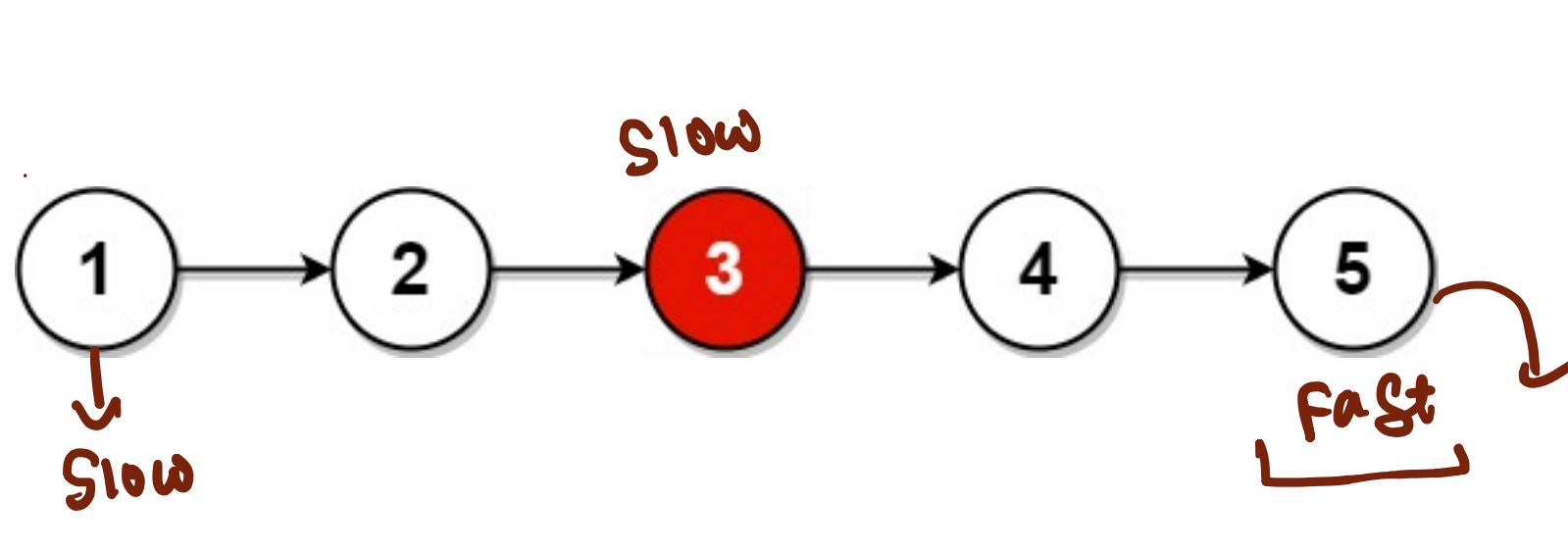
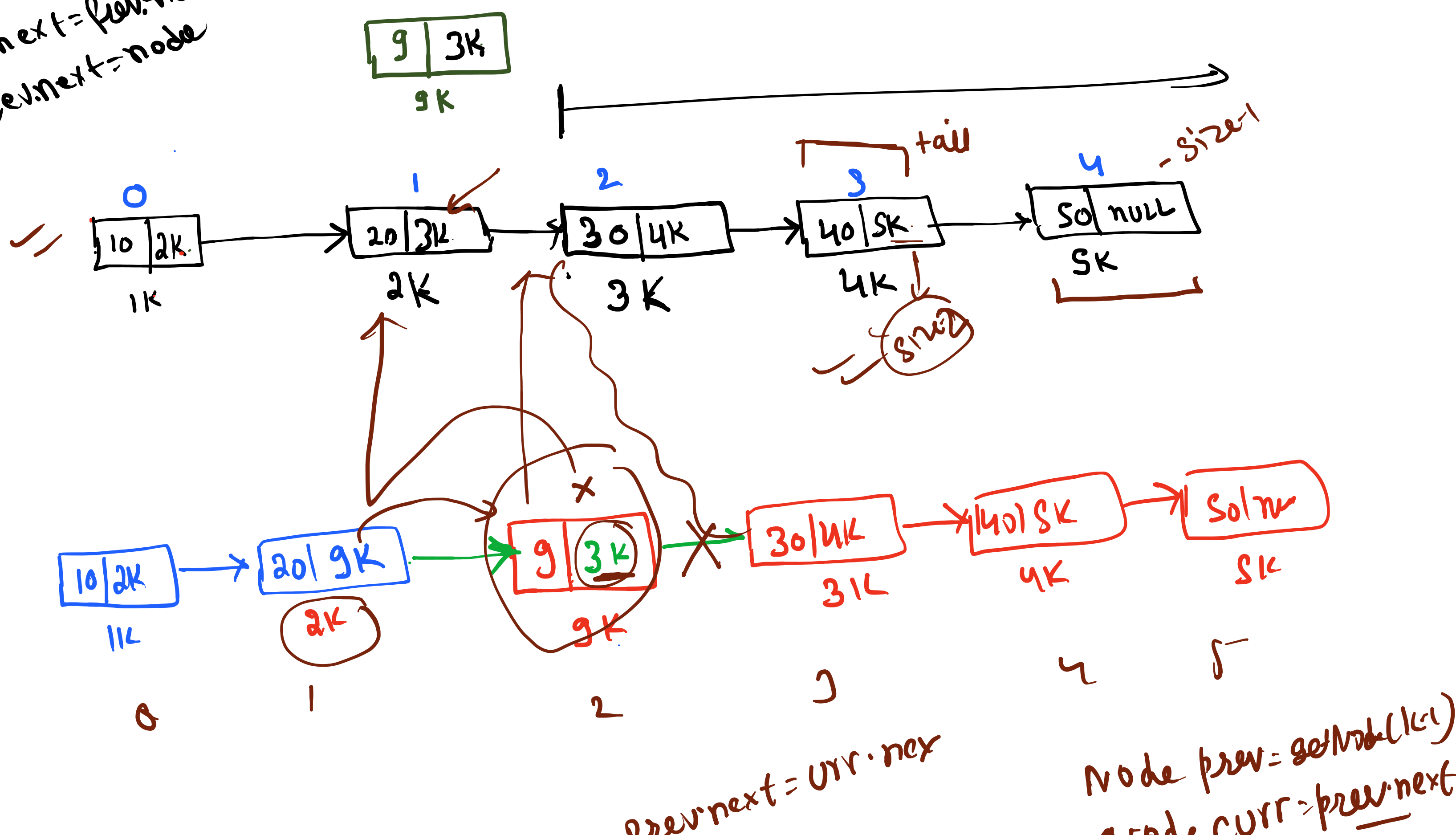
```
Node temp = head;
for (i = 1; i <= k; i++) {
    temp = temp.next;
}
```

10 20 30 40 50

Node temp = head

```
while (head != null) {
    System.out.println(head.val);
    head = head.next;
}
```

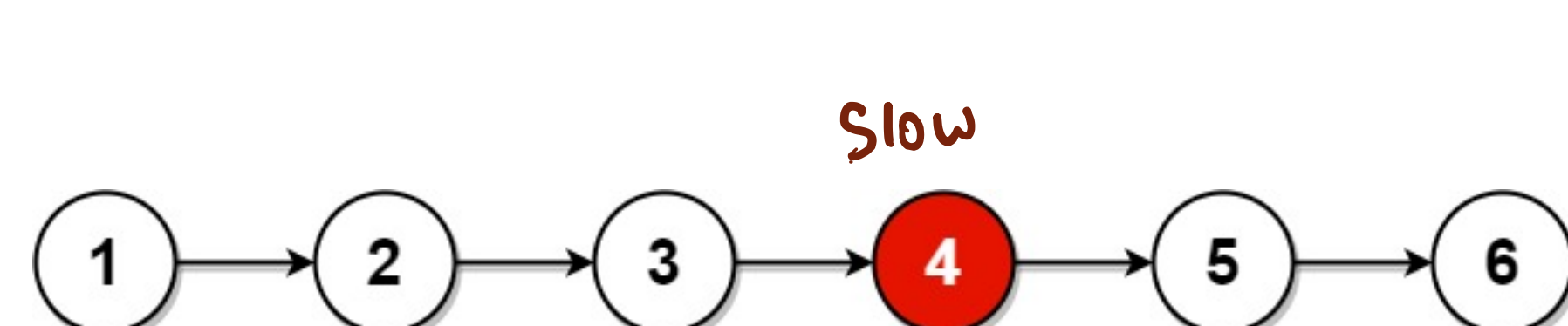
node.next = prev.next
prev.next = node



while (fast != null && fast.next != null) {

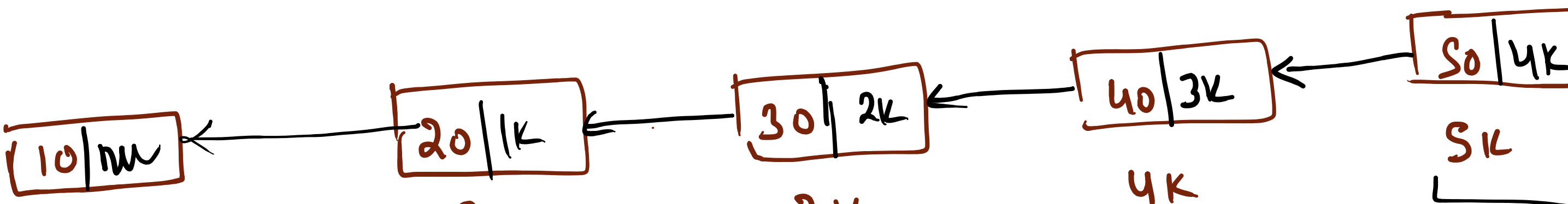
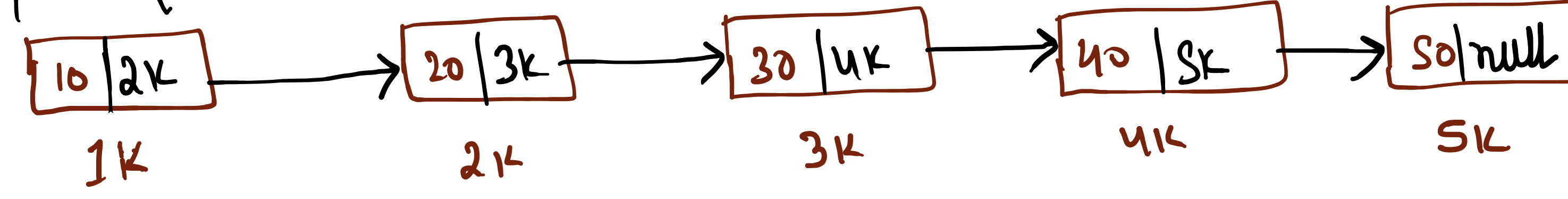
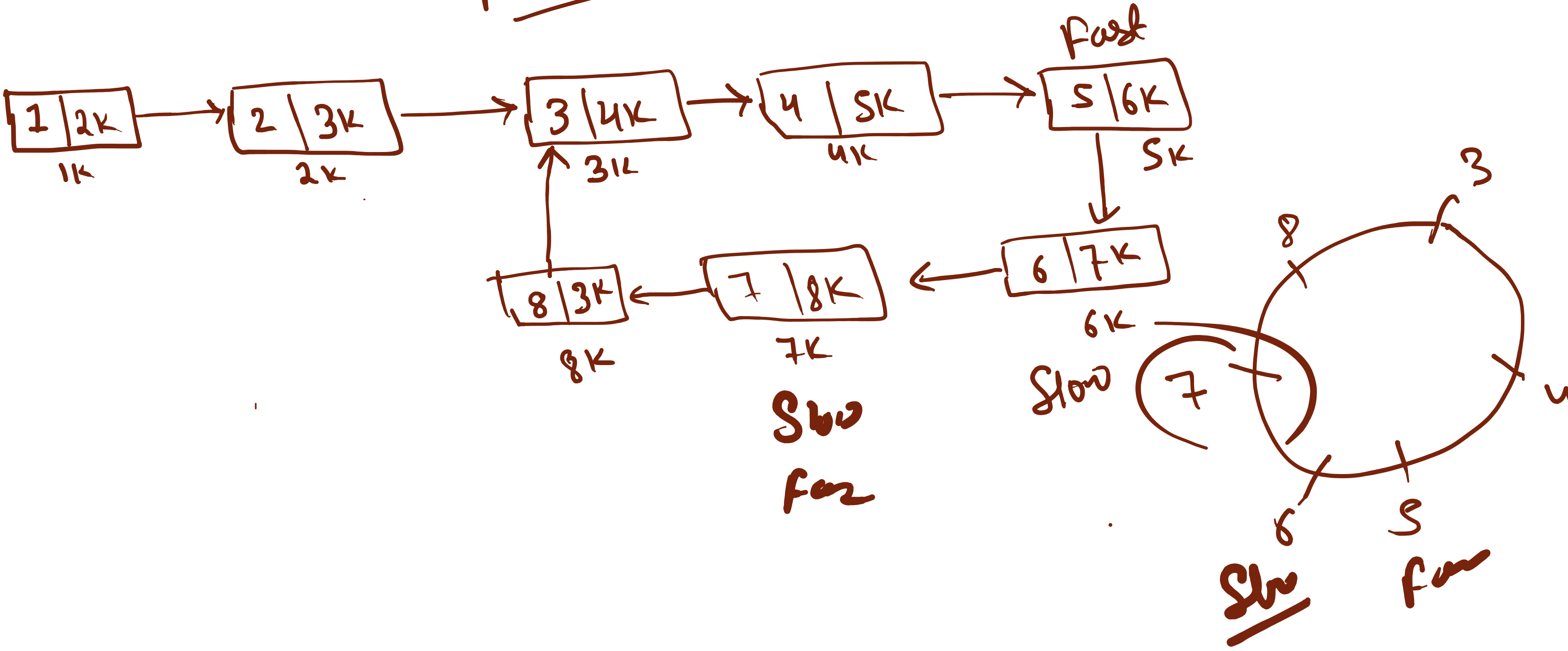
slow = slow.next;

fast = fast.next.next;



fast != null

Floyd cycle detection



Node curr = head
Node prev = null

ahead = 2K

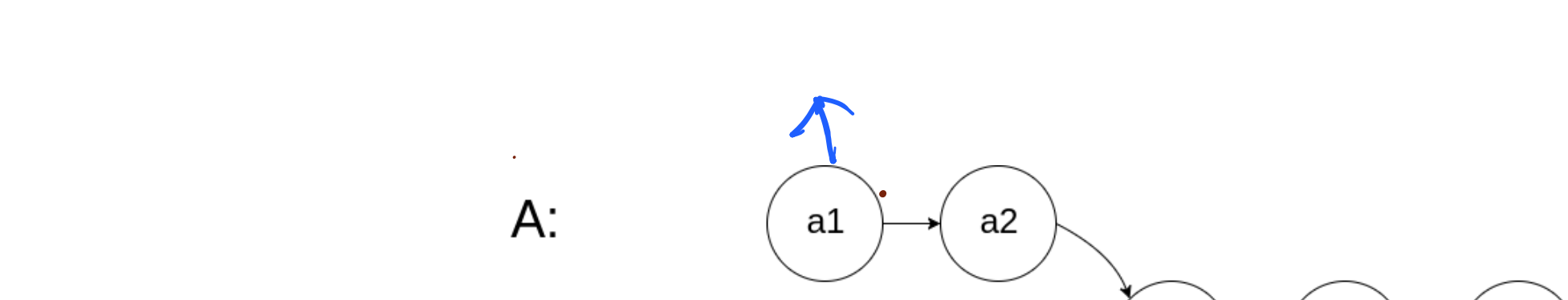
while (curr != null) {

Node ahead = curr.next;

curr.next = prev;

prev = curr;

curr = ahead;



8 - 5 = 3

len x

